

```

tarun@LAPTOP-BRD2JBFC:~$ cd ~/oslab/2520090020_Practical/practical/
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical/practical/$ ^C
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical/practical/$ cat /proc/1909/maps
628b714f6000-628b714f7000 r--p 00000000 08:30 42872 /home/tarun/oslab/2520090020_Practical/practical7/memory
628b714f7000-628b714f8000 r--p 00001000 08:30 42872 /home/tarun/oslab/2520090020_Practical/practical7/memory
628b714f8000-628b714f9000 r--p 00002000 08:30 42872 /home/tarun/oslab/2520090020_Practical/practical7/memory
628b714f9000-628b714fa000 r--p 00003000 08:30 42872 /home/tarun/oslab/2520090020_Practical/practical7/memory
628b714fa000-628b714fb000 rw-p 00003000 08:30 42872 /home/tarun/oslab/2520090020_Practical/practical7/memory
628b96cfd000-628b96d1e000 rw-p 00000000 00:00 0 [heap]
729f67400000-729f67428000 r--p 00000000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
729f67428000-729f675c0000 r--p 00028000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
729f675c0000-729f6760e000 r--p 001c0000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
729f6760e000-729f67612000 r--p 0020d000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
729f67612000-729f67614000 rw-p 00211000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
729f67614000-729f67621000 rw-p 00000000 00:00 0
729f676c6000-729f676c9000 rw-p 00000000 00:00 0
729f676d1000-729f676d3000 rw-p 00000000 00:00 0
729f676d3000-729f676d7000 r--p 00000000 00:00 0 [vvar]
729f676d7000-729f676d9000 r--p 00000000 00:00 0 [vvar_vclock]
729f676d9000-729f676db000 r--p 00000000 00:00 0 [vdso]
729f676db000-729f676dc000 r--p 00000000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
729f676dc000-729f6770b000 r--p 00001000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
729f6770b000-729f67716000 r--p 00030000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
729f67716000-729f67718000 r--p 0003b000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
729f67718000-729f67719000 rw-p 0003d000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
729f67719000-729f6771a000 rw-p 00000000 00:00 0
7fff5b8f000-7fff5b8e1000 rw-p 00000000 00:00 0 [stack]
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical/practical7$

```

```

tarun@LAPTOP-BRD2JBFC:~$ cd ~/oslab/2520090020_Practical
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical$ mkdir practical7
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical$ cd practical7
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical/practical7$ pwd
/home/tarun/oslab/2520090020_Practical/practical7
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical/practical7$ gedit memory.c

```

```

** (gedit:614): WARNING **: 10:14:21.246: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

```

```

** (gedit:614): WARNING **: 10:14:21.246: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
^C

```

```

tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical/practical7$ gcc -Wall -g memory.c -o memory
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical/practical7$ ./memory
Process ID: 1909

```

```

Memory addresses:
Code : 0x628b714f7229
Global : 0x628b714fa010
Static : 0x628b714fa014
Static local : 0x628b714fa018
Heap : 0x628b96cfd010
Stack : 0x7fff5b8df68c

```

```

Process is running...
Use another terminal to check /proc/1909/maps
Press Enter to exit...

```

```

tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical/practical7$ cat /proc/1909/maps | grep -E "heap|stack|memory"
628b714f6000-628b714f7000 r--p 00000000 08:30 42072 /home/tarun/oslab/2520090020_Practical/practical7/memory
628b714f7000-628b714f8000 r-xp 00001000 08:30 42072 /home/tarun/oslab/2520090020_Practical/practical7/memory
628b714f8000-628b714f9000 r--p 00002000 08:30 42072 /home/tarun/oslab/2520090020_Practical/practical7/memory
628b714f9000-628b714fa000 r--p 00003000 08:30 42072 /home/tarun/oslab/2520090020_Practical/practical7/memory
628b714fa000-628b714fb000 rw-p 00003000 08:30 42072 /home/tarun/oslab/2520090020_Practical/practical7/memory
628b96cfd000-628b96d1e000 rw-p 00000000 00:00 0 [heap]
7fff5b8bf000-7fff5b8e1000 rw-p 00000000 00:00 0 [stack]
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical/practical7$ grep -E "heap|stack" /proc/1909/maps
628b96cfd000-628b96d1e000 rw-p 00000000 00:00 0 [heap]
7fff5b8bf000-7fff5b8e1000 rw-p 00000000 00:00 0 [stack]

```

```

tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical/practical7$ pmap 904
904: ./memory
00005c2794f55000      4K r---- memory
00005c2794f56000      4K r-x-- memory
00005c2794f57000      4K r---- memory
00005c2794f58000      4K r---- memory
00005c2794f59000      4K rw--- memory
00005c27c437e000    132K rw--- [ anon ]
0000712a12e00000    160K r---- libc.so.6
0000712a12e28000   1632K r-x-- libc.so.6
0000712a12fc0000    312K r---- libc.so.6
0000712a1300e000     16K r---- libc.so.6
0000712a13012000      8K rw--- libc.so.6
0000712a13014000     52K rw--- [ anon ]
0000712a131bb000     12K rw--- [ anon ]
0000712a131c6000      8K rw--- [ anon ]
0000712a131c8000     16K r---- [ anon ]
0000712a131cc000      8K r---- [ anon ]
0000712a131ce000      8K r-x-- [ anon ]
0000712a131d0000      4K r---- ld-linux-x86-64.so.2
0000712a131d1000   188K r-x-- ld-linux-x86-64.so.2
0000712a13200000    44K r---- ld-linux-x86-64.so.2
0000712a1320b000      8K r---- ld-linux-x86-64.so.2
0000712a1320d000      4K rw--- ld-linux-x86-64.so.2
0000712a1320e000      4K rw--- [ anon ]
00007ffe452ad000   136K rw--- [ stack ]
total                2772K

```

```

tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical/practical7$ cat /proc/904
/status | grep -E "VmSize|VmRSS|VmData|VmStk"
VmSize:      2772 kB
VmRSS:       1808 kB
VmData:       224 kB
VmStk:       136 kB
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical/practical7$ |

```